Shared Structure Without Stronger Scale Dependence: Qualifying the Platonic Scaling Prediction in Cross-Modal Astronomy

Anonymous authors

Abstract

The Platonic Representation Hypothesis predicts increasing representational convergence with model scale [Huh et al., 2024]. We ask whether that relationship becomes clearer when cross-modal correspondence is measured through substantially different probes on matched Legacy Survey–HSC embeddings [UniverseTBD et al., 2025]. Shared sparse representations recover substantially more held-out neighbourhood correspondence than the dense baseline, but the additional alignment they expose shows little systematic dependence on model size. Within-family probe×scale interactions are small and mixed in sign (SAE mean $D{=}{-}5.2{\times}10^{-4}$; BSF mean $D{\approx}0$; 6/11 positive in each case). A nonlinear alignment trained without paired objects [Jha et al., 2025] also recovers nontrivial relational geometry, yet its size dependence is non-monotonic. So the amount of shared structure we measure depends strongly on the probe, while the way that structure changes with raw model size is much less consistent. This makes raw parameter count look like, at best, an incomplete control variable for representational convergence.

1 Introduction

Representations from independently trained models often become more aligned as capability and scale increase [Huh et al., 2024]. The Platonic Representation Hypothesis (PRH) interprets this as convergence toward a shared statistical model of reality [Huh et al., 2024]. In astronomy, matched survey embeddings give a relatively clean test of this idea: previous work reports increasing mutual k-nearest-neighbour (mKNN) alignment with model capacity [UniverseTBD et al., 2025].

There is an immediate ambiguity, however. Measured alignment depends on how the representation is read. If a structured probe exposes substantially more Legacy↔HSC correspondence than the dense embeddings do, does that extra correspondence appear preferentially in the larger models?

We test that question directly. On a matched held-out evaluation, shared-basis sparse autoencoders (SAEs) [Gao et al., 2025, Lan et al., 2024] and block-sparse featurizers (BSFs) [Fel et al., 2026] increase neighbourhood overlap across every model-size rung. We define the lift

$$L_R(P) = M_R(P) - M_{\mathrm{dense}}(P),$$

and ask whether $L_R(P)$ itself grows with P. It does not in any clear or systematic way. The corresponding within-family interaction,

$$D_R = \Delta M_R - \Delta M_{\mathrm{dense}},$$

is small and mixed in sign.

We then repeat the question with a qualitatively different probe: an unpaired nonlinear alignment inspired by Jha et al. [2025]. It recovers nontrivial relational structure without seeing paired training objects, but again gives no clean monotonic relationship with model size.

The point is therefore not that any particular probe "fails to scale." It is that changing the probe changes the *level* of measured correspondence much more reliably than it changes the *size response*. That distinction is the main result of the paper.

2 Experimental setup and alignment probes

2.1 Legacy↔HSC model-size ladders

We use the official UniverseTBD Legacy Survey [Dey et al., 2019]↔HSC [Miyazaki et al., 2018] matched embeddings [UniverseTBD et al., 2025]. Each rung pairs the same architecture size across surveys; we use n=16,384 matched objects. The model families are AstroPTv2 (15, 95, 850M) [Smith et al., 2024], ConvNeXtv2 (nano–large) [Woo et al., 2023], DINOv2 (small–giant) [Oquab et al., 2024], ViT (base–huge) [Dosovitskiy et al., 2021], and I-JEPA (huge, giant) [Assran et al., 2023]. The unpaired analysis covers ConvNeXt and DINOv2.

2.2 mKNN and matched evaluation

For paired representations A and B,

$$M_k = \frac{1}{n}\sum_i \frac{|\mathcal{N}_k^A(i) \cap \mathcal{N}_k^B(i)|}{k},$$

using cosine neighbours with self-exclusion [Huh et al., 2024, UniverseTBD et al., 2025]. Neighbour lists are sliced to k before intersection. Our primary scale is k=10. Under a uniform-neighbour null, expected overlap is $k/(n-1)$ [Gröger et al., 2026].

We retain the original full-catalog dense score only to connect back to the scaling analysis of UniverseTBD et al. [2025]. All probe×scale comparisons use the same held-out query IDs, full gallery, k, self-exclusion, and object set (n_{test}=3277). Dense, SAE, and BSF are therefore directly comparable in both level and size response.

2.3 Dense, SAE, and BSF probes

The dense probe uses the raw ℓ_2-normalised embeddings with cosine mKNN.

For SAE, we encode both surveys with pretrained TopK sparse autoencoders [Gao et al., 2025], then fit a Ridge map from Legacy codes into the HSC basis using train IDs only, with train-only IDF weighting (`shared_side1_basis_idf`). The mapping direction is fixed in advance. Dictionary width F=2048 and seed 0 are shared across the ladder; TopK sparsity varies with the available checkpoint ($k \in \{18, \ldots, 23\}$).

For BSF, we replace SAE codes with Vanilla BSF full signed codes [Fel et al., 2026] (G≈512 blocks) and apply the same shared-basis protocol.

For each within-family adjacent step $P_j \to P_{j+1}$, we compute

$$D_{R,j} = \Delta M_{R,j} - \Delta M_{\text{dense},j} = L_R(P_{j+1}) - L_R(P_j),$$

for $R \in \{\text{SAE}, \text{BSF}\}$. This is the quantity we care about: whether the extra correspondence exposed by the structured probe itself increases with size.

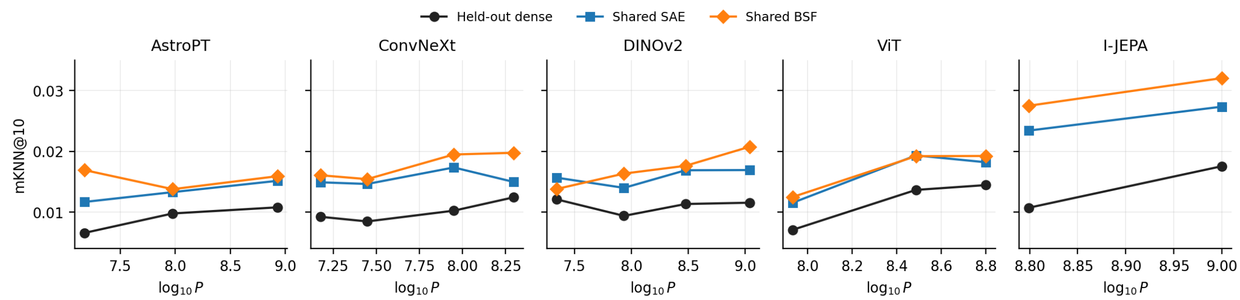

Figure 1: Held-out mKNN@10 across Legacy↔HSC model ladders using identical queries and galleries for dense, shared SAE, and shared BSF. Probe choice changes the level of measured correspondence substantially more than the within-family size trajectory.

2.4 Unpaired relational probe

Our second analysis uses a DualEncoder inspired by Jha et al. [2025]. Each survey is mapped through a model-specific MLP into a shared latent Z=256 and decoded back. Training combines reconstruction, RBF-MMD, cycle consistency, and Gram-matrix geometry preservation with equal weights (80 epochs, hidden width 512, two seeds). The two training sets are disjoint (5500 objects per survey), so the model never sees paired cross-survey correspondences.

Evaluation uses a held-out paired set (n=2884) and measures neighbourhood agreement between the translated representation and the true opposite-survey embedding. This is a different relational functional from the matched SAE/BSF comparison, so we use it to test the *shape* of the size dependence rather than to rank absolute mKNN values across methods.

3 Results

3.1 Structured probes reveal substantially more correspondence

Against the matched held-out dense baseline at k=10, SAE lifts every rung (mean +0.0056, range +0.0025–+0.0127), while BSF gives a larger mean lift of +0.0076 (range +0.0017–+0.0168). For example, ConvNeXt nano moves from 0.0092 dense to 0.0149 with SAE and 0.0161 with BSF; I-JEPA giant moves from 0.0176 to 0.0274 and 0.0321, respectively. The dense paper-style scores are already above chance ($\approx$ 0.007–0.018, mean 0.0115), but the structured probes expose substantially more correspondence.

Figure 1 shows the important visual pattern: the probes produce large vertical shifts, while the within-family trajectories change much less.

3.2 The extra correspondence does not strengthen systematically with scale

Under the original paper-style dense evaluation, 8/11 adjacent size steps are positive (one-sided p=0.113), which recovers the mild direction of the earlier astronomy scaling result on this Legacy-only subset. The main analysis, however, is the matched held-out comparison.

The 11 within-family probe×scale interactions cluster around zero (Figure 2b). For SAE, mean D=-5.2×10^{-4}, median $+4.9\times10^{-4}$, range $[-4.6, +1.2]\times10^{-3}$, with 6 positive and 5 negative. For BSF, mean D≈0, median $+1.2\times10^{-4}$, range $[-6.3, +5.2]\times10^{-3}$, again with 6 positive and 5 negative.

The same picture appears in the lift itself. Spearman correlation between $L_R(P)$ and $\log_{10} P$ is +0.11 for SAE and +0.13 for BSF. In other words, the structured probes expose more shared

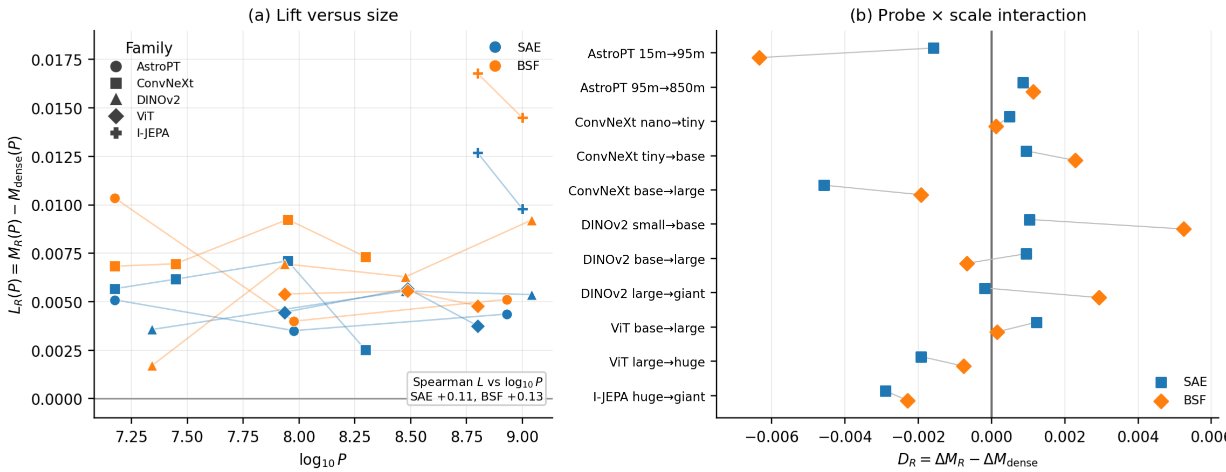

Figure 2: (a) Matched held-out lift $L_R(P)$ versus $\log_{10} P$. (b) Adjacent probe×scale interaction $D_R = \Delta M_R - \Delta M_{\mathrm{dense}}$ for the same 11 within-family steps. The lift remains positive across the ladder, while its change with model size is small and mixed in sign.

structure, but that extra structure does not preferentially appear in the larger models.

Raw adjacent-sign counts are consistent with this reading rather than being the main result. On the matched holdout they are 9/11 for dense, 7/11 for SAE, and 9/11 for BSF. This is why it would be misleading to say that BSF "does not scale": it can rise with size while its *increment over dense* remains approximately size-independent.

3.3 Unpaired alignment gives the same qualitative size picture

The unpaired probe also recovers nontrivial relational structure. Its mKNN@10 scores lie around 0.012–0.023, compared with a random-neighbour floor of $10/2883 \approx 0.0035$, despite the model never seeing paired cross-survey objects during training. Its size trajectories are nevertheless non-monotonic.

For ConvNeXt, mKNN@10 follows

$$0.016 \to 0.012 \to 0.023 \to 0.012,$$

while DINOv2 gives

$$0.022 \to 0.019 \to 0.020 \to 0.019.$$

Only 2/6 adjacent steps are positive. With six transitions, the trajectory itself is more informative than a significance test.

This does not put unpaired alignment on the same absolute scale as SAE or BSF. Its value is that the absence of clean monotonicity persists under a substantially different alignment construction.

4 Discussion and limitations

The main result is a separation between alignment magnitude and scale dependence. SAE and BSF expose substantially more Legacy↔HSC correspondence than dense embeddings, yet the corresponding probe×scale interactions remain small and heterogeneous. The unpaired result points

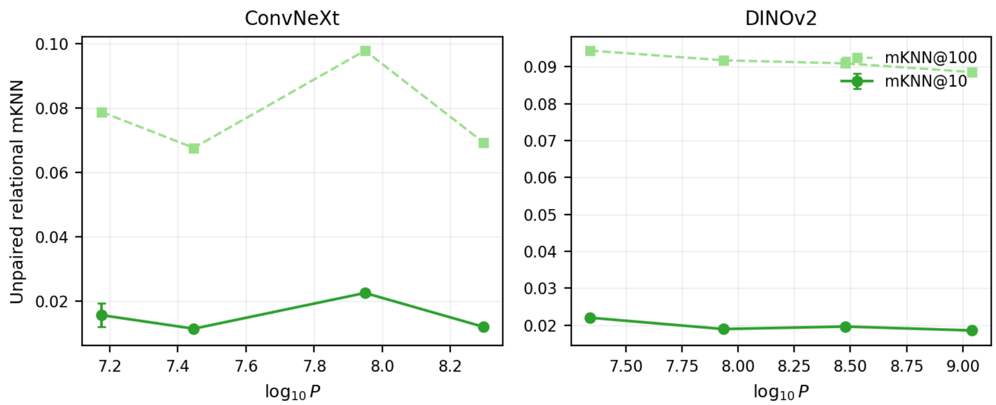

Figure 3: Unpaired DualEncoder relational mKNN versus $\log_{10} P$ (mean $\pm$ seed sd; two seeds). Solid: mKNN@10. Dashed: mKNN@100. The curves are non-monotonic across both available families.

in the same direction from a different construction. Taken together, this weakens the idea that the observed size response is mainly an artefact of reading the representations in dense coordinates.

One possibility is that raw parameter count is simply not the right variable. It is not an invariant measure of functional complexity: parameters can be added redundantly without materially changing the represented function. Model size may therefore be standing in for something more relevant, such as the pressure to achieve capability under constrained computational resources. Under that view, representational convergence could depend more directly on efficiency or compression pressure than on parameter count itself. We treat this as a hypothesis for future work. The connection to computationally relative notions of structure such as epiplexity [Finzi et al., 2026] is suggestive, but it is not needed for the present result.

The limitations are straightforward. There are only 11 paired adjacent transitions, and the unpaired analysis covers two families. SAE TopK sparsity is not perfectly homogeneous across the ladder ($k \in \{18, \ldots, 23\}$, with fixed F and seed). The unpaired analysis uses a different training and evaluation construction by design. Finally, learned alignment procedures can induce correspondence in ways that an analytical mKNN chance baseline does not capture; pipeline-refit permutation nulls are therefore a natural next control [Gröger et al., 2026].

5 Conclusion

Structured probes substantially increase measured cross-modal correspondence, but the additional correspondence they expose does not systematically increase with model size. A qualitatively different unpaired relational probe gives the same broad picture over the available ladders. Probe choice strongly affects how much shared structure is visible; its interaction with raw model scale is considerably weaker and less consistent.

The next question, then, is not simply whether larger models become more Platonic. It is what quantity actually governs representational convergence.

Acknowledgments

Code and exact configurations will be released with the camera-ready version.

References

Mahmoud Assran, Quentin Duval, Ishan Misra, Piotr Bojanowski, Pascal Vincent, Michael Rabbat, Yann LeCun, and Nicolas Ballas. Self-supervised learning from images with a joint-embedding predictive architecture. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 15619–15629, 2023. doi: 10.1109/CVPR52729.2023.01499.

Arjun Dey, David J. Schlegel, Dustin Lang, Robert Blum, Kaylan Burleigh, Xiaohui Fan, Joseph R. Findlay, Doug Finkbeiner, David Herrera, Stéphanie Juneau, et al. Overview of the DESI legacy imaging surveys. *The Astronomical Journal*, 157(5):168, 2019. doi: 10.3847/1538-3881/ab089d.

Alexey Dosovitskiy, Lucas Beyer, Alexander Kolesnikov, Dirk Weissenborn, Xiaohua Zhai, Thomas Unterthiner, Mostafa Dehghani, Matthias Minderer, Georg Heigold, Sylvain Gelly, Jakob Uszkoreit, and Neil Houlsby. An image is worth 16x16 words: Transformers for image recognition at scale. In *International Conference on Learning Representations*, 2021.

Thomas Fel, Matthew Kowal, Mozes Jacobs, Dron Hazra, Usha Bhalla, Lee Sharkey, Lucius Bushnaq, Satchel Grant, Tal Haklay, Thomas Icard, Can Rager, Michael Pearce, Daniel Wurgaft, Aiden Swann, Fenil Doshi, Siddharth Boppana, Curt Tigges, Nick Cammarata, Thomas Serre, Vasudev Shyam, Owen Lewis, Thomas McGrath, Jack Merullo, Ekdeep Singh Lubana, and Atticus Geiger. Structuring sparsity: Block-sparse featurizers capture visual concept manifolds. *arXiv preprint arXiv:2606.25234*, 2026.

Marc Finzi, Shikai Qiu, Yiding Jiang, Pavel Izmailov, J. Zico Kolter, and Andrew Gordon Wilson. From entropy to epiplexity: Rethinking information for computationally bounded intelligence. *arXiv preprint arXiv:2601.03220*, 2026.

Leo Gao, Tom Dupré la Tour, Henk Tillman, Gabriel Goh, Rajan Troll, Alec Radford, Ilya Sutskever, Jan Leike, and Jeffrey Wu. Scaling and evaluating sparse autoencoders. In *International Conference on Learning Representations*, 2025. URL `https://arxiv.org/abs/2406.04093`.

Fabian Gröger, Shuo Wen, and Maria Brbić. Revisiting the Platonic representation hypothesis: An Aristotelian view. *arXiv preprint arXiv:2602.14486*, 2026.

Minyoung Huh, Brian Cheung, Tongzhou Wang, and Phillip Isola. Position: The Platonic representation hypothesis. In Ruslan Salakhutdinov, Zico Kolter, Katherine Heller, Adrian Weller, Nuria Oliver, Jonathan Scarlett, and Felix Berkenkamp, editors, *Proceedings of the 41st International Conference on Machine Learning*, volume 235 of *Proceedings of Machine Learning Research*, pages 20617–20642. PMLR, 2024. URL `https://proceedings.mlr.press/v235/huh24a.html`.

Rishi Jha, Collin Zhang, Vitaly Shmatikov, and John X. Morris. Harnessing the universal geometry of embeddings. In *Advances in Neural Information Processing Systems*, 2025. URL `https://proceedings.neurips.cc/paper_files/paper/2025/hash/4175dee33d6145cb8f0323703d138a53-Abstract-Conference.html`.

Michael Lan, Philip Torr, Austin Meek, Ashkan Khakzar, David Krueger, and Fazl Barez. Quantifying feature space universality across large language models via sparse autoencoders. *arXiv preprint arXiv:2410.06981*, 2024.

Satoshi Miyazaki, Yutaka Komiyama, Satoshi Kawanomoto, Yukiko Fujimori, Takashi Hamana, Hisanori Hayashi, Hiroki Heiguchi, Takashi Hattori, Shiang-Yu Huang, Masatoshi Imanishi, et al. Hyper Suprime-Cam: System design and verification of image quality. *Publications of the Astronomical Society of Japan*, 70(SP1):S1, 2018. doi: 10.1093/pasj/psx063.

Maxime Oquab, Timothée Darcet, Théo Moutakanni, Huy V. Vo, Marc Szafraniec, Vasil Khalidov, Pierre Fernandez, Daniel Haziza, Francisco Massa, Alaaeldin El-Nouby, Mahmoud Assran, Nicolas Ballas, Wojciech Galuba, Russell Howes, Po-Yao Huang, Shang-Wen Li, Ishan Misra, Michael Rabbat, Vasu Sharma, Gabriel Synnaeve, Hu Xu, Hervé Jégou, Julien Mairal, Patrick Labatut, Armand Joulin, and Piotr Bojanowski. DINOv2: Learning robust visual features without supervision. *Transactions on Machine Learning Research*, 2024.

Michael J. Smith, Ryan J. Roberts, Eirini Angeloudi, and Marc Huertas-Company. AstroPT: Scaling large observation models for astronomy. In *ICML 2024 AI for Science Workshop*, 2024.

UniverseTBD, Kshitij Duraphe, Michael J. Smith, Shashwat Sourav, and John F. Wu. The Platonic universe: Do foundation models see the same sky? *arXiv preprint arXiv:2509.19453*, 2025. doi: 10.48550/arXiv.2509.19453. Accepted to the NeurIPS 2025 Workshop on Machine Learning and the Physical Sciences.

Sanghyun Woo, Shoubhik Debnath, Ronghang Hu, Xinlei Chen, Zhuang Liu, In So Kweon, and Saining Xie. ConvNeXt V2: Co-designing and scaling ConvNets with masked autoencoders. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 16133–16142, 2023.